NyaySaakshya

Cryptographic Security Posture Assessment

Assessment 2647f6ab59fca5e9
Capture c2a84bf4ed973fcdff29bc5dd8ea24791d195fa186be3bbe7a359f68a3201fff
Run 33fbadb0e55244aa8a05eee2de96c9f0
Analysed 2026-10-03T04:36:38Z

OVERALL POSTURE	POSTURE SCORE	EVIDENCE COVERAGE	ML LANE
ADEQUATE	88 / 100	100.0% (2 of 2 sessions assessed)	Disabled

1. Executive summary

Overall posture	ADEQUATE
Posture score	88 / 100 F2-group-damped
Evidence coverage	100.0% (2 of 2 sessions assessed)
Protocols observed	smtp
Issue groups	7 total · 1 penalising the score
Abstentions	7
ML lane	Disabled

2 session(s) were reconstructed from the capture; 2 were assessed and 0 were abstained from.

1 issue group(s) penalise the posture score. The highest severity recorded is MEDIUM.

2. Assessment scope

What this assessment covers, and what it structurally cannot.

Analysis type	Passive PCAP analysis
Capture identifier (SHA-256)	c2a84bf4ed973fcdff29bc5dd8ea24791d195fa186be3bbe7a359f68a3201fff
Analysis timestamp	2026-10-03T04:36:38Z time of analysis, not of report generation

NyaySaakshya observes traffic that was already captured. It does not connect to any server, hold any private key, decrypt any session, or read message content. Anything that cannot be determined from the captured bytes is reported as not observable rather than inferred.

Sessions per protocol

PROTOCOL	SESSIONS
smtp	2

3. Overall security posture

Posture	ADEQUATE
Score	88 / 100
Evidence coverage	100.0% (2 of 2 sessions assessed)
Scoring formula	F2-group-damped

100 minus 12.00 across 1 issue group(s) under F2-group-damped

The score starts at 100 and each penalty above is subtracted from it. Only observed issues penalise; compliant observations earn no credit, and missing evidence neither penalises nor rewards.

Score decomposition

ISSUE CLASS	SEVERITY	RECURRENCE	WEIGHT	RECURRENCE MULTIPLIER	PENALTY
No tls protection	MEDIUM	1	12	1	-12

The severity weights and band thresholds behind this score are transparent engineering policy, not a calibrated measurement derived from incident data.

4. Evidence coverage

How much of the captured traffic the analysis could actually assess.

Sessions total	2
Sessions assessed	2
Sessions abstained	0
Assessed fraction	100.0%

Coverage is reported beside the posture score and is never folded into it. A score computed over a small share of the traffic and the same score computed over nearly all of it are different claims.

Evidence states observed across all fields

EVIDENCE STATE	COUNT	SHARE
OBSERVED	12	75.0%
UNKNOWN	4	25.0%

Session completeness

COMPLETENESS	SESSIONS
COMPLETE	1
INCOMPLETE	1

5. Protocol posture

Posture per protocol, across the dimensions the evidence supports.

Per-protocol posture

PROTOCOL	SESSIONS	BAND	SCORE	DIMENSIONS ASSESSED	NOT OBSERVABLE	ABSTENTIONS
smtp	2	ADEQUATE	88	CRYPTO CONFIGURATION · PLAINTEXT EXPOSURE · PROTOCOL VERSION · UPGRADE INTEGRITY	—	7

6. Prioritised findings

Investigative priority, as ranked by the assessment. This ordering is not a severity ranking.

Severity states how serious a condition is. Priority states the order in which an analyst should work. The ML column shows any bounded adjustment the anomaly lane contributed to the ordering; it cannot move a finding across a severity tier and it never changes a severity.

Findings in investigative priority order

RANK	SEVERITY	ISSUE	STATUS	CERTAINTY	SESSIONS	PRIORITY	ML ADJ.
1	[!] MEDIUM	No tls protection	OBSERVED ISSUE	CONFIRMED	1	56	0

7. Detailed findings

Mail session carried no TLS — The mail session carried no TLS protection.

No TLS records were observed in this stream, so message and command content travelled in cleartext. This states what was observed. It does not attribute a cause: a session may be unprotected through client choice, server configuration, or interference, and this capture alone cannot distinguish those.

Recurrence counts the sessions in which a condition was observed. Findings are grouped by condition, so one misconfiguration seen many times is one issue with a recurrence count, not many issues.

Issue groups with evidence and standards basis

SEVERITY	ISSUE	FACT KIND	DIMENSION	CERTAINTY	OBSERVABILITY	RECURRENCE	STANDARD	FRAMES	PENALISING
[!] MEDIUM	Mail session carried no TLS	Base security issue	Plaintext exposure	CONFIRMED	OBSERVABLE	1	NIST SP 800-52r2 · RFC 8314	—	Yes
[i] INFO	Insecure protocol configuration	Positive evidence	Crypto configuration	UNDETERMINED	—	2	NIST SP 800-52r2 · RFC 8996 (BCP 195) · UNMAPPED · UNMAPPED	—	No
[i] INFO	Negotiated TLS version posture	Positive evidence	Protocol version	CONFIRMED	—	1	NIST SP 800-52r2 · RFC 8996 (BCP 195)	—	No
[i] INFO	Forward secrecy	Positive evidence	Crypto configuration	PROBABLE	—	1	NIST SP 800-52r2 · UNMAPPED	—	No
[i] INFO	Key exchange mechanism	Positive evidence	Crypto configuration	CONFIRMED	—	1	UNMAPPED	—	No
[i] INFO	STARTTLS/STLS upgrade outcome	Positive evidence	Upgrade integrity	CONFIRMED	—	1	RFC 3207	—	No
[i] INFO	TLS handshake completion evidence	Positive evidence	Crypto configuration	CONFIRMED	—	1	UNMAPPED	—	No

8. Standards basis

Standards cited by the rules that produced these findings.

Standards cited in this assessment

STANDARD	CITATIONS	SECTIONS
distinct_standards	4	—
note	every citation originates from a Phase-4/5 rule; the Phase-7 registry adds structure and introduces no new standard	—

A standard appearing here means a rule cited it as the basis for an observation. It is not a statement of compliance or certification against that standard.

9. Remediation guidance

Rule-bound guidance carried from the assessment, in its own words.

Recommended actions

OBSERVED	WHY IT MATTERS	RECOMMENDED ACTION	AFFECTED SCOPE	VERIFICATION	STANDARD
A mail session completed without any TLS protection.	The entire session, including message content and any credentials, was carried in cleartext. RFC 8314 declares this obsolete for submission and access.	Require TLS for this service, preferring implicit TLS on the dedicated port, or enforce STARTTLS/STLS for the cleartext port. Rule guidance: Require TLS for this service, preferring implicit TLS on the dedicated ports (RFC 8314) or enforced STARTTLS/STLS.	session c2a84bf4ed973fcdff29bc5dd8ea24791d195fa186be3bbe7a359f68a3201fff:0 (smtp)	Re-capture and confirm TLS records are present on every session to this service.	NIST SP 800-52r2 · RFC 8314

this recommendation addresses what the capture showed; the service may have other configuration that was not exercised by the captured traffic

NyaySaakshya cannot confirm that a change was applied or effective -- only a later capture can

This system does not verify that remediation was performed or that it succeeded. Each recommendation above includes its own verification step, which must be carried out and confirmed independently.

10. AI / ML transparency

What the machine-learning lane did, and what it is not permitted to do.

ML lane	Disabled

ML lane disabled; posture computed from deterministic and cross-session evidence only

11. Abstentions and observability limits

Questions this assessment declined to answer, and what would settle them.

Abstentions

REASON	ISSUE CLASS	COULD NOT CONCLUDE	WHY	WOULD BE RESOLVED BY	PROTOCOL	FRAMES
Not observable	Certificate extraction	Certificate extraction	No certificate was observable in this session.	not resolvable by passive capture; this fact is structurally hidden from an observer without key material	smtp	14
Not observable	Certificate observability	Certificate trust and revocation boundary	Certificate trust and revocation were not evaluated.	not resolvable by passive capture; this fact is structurally hidden from an observer without key material	smtp	14
Insufficient capture	Starttls advertisement	STARTTLS/STLS advertisement posture	No capability response was captured for this session.	re-capture the full session including connection setup and teardown	smtp	—
Insufficient capture	Starttls behaviour deviation	STARTTLS/STLS advertisement deviation from baseline	Insufficient comparable history to assess this session.	re-capture the full session including connection setup and teardown	smtp	—
Insufficient capture	Starttls behaviour deviation	STARTTLS/STLS upgrade deviation from baseline	Insufficient comparable history to assess this session.	re-capture the full session including connection setup and teardown	smtp	—
Insufficient capture	Starttls behaviour deviation	STARTTLS/STLS advertisement deviation from baseline	Insufficient comparable history to assess this session.	re-capture the full session including connection setup and teardown	smtp	10
Insufficient capture	Starttls behaviour deviation	STARTTLS/STLS upgrade deviation from baseline	Insufficient comparable history to assess this session.	re-capture the full session including connection setup and teardown	smtp	10

An abstention is neither a failure nor a pass. It records that the evidence did not support a conclusion. Treating any row above as compliant, or as a detected problem, misreads it.

12. Assessment limitations

Constraints that bound what this assessment can support.

passive analysis: conclusions describe what the capture shows, not the server's configuration

no attacker, intent or attribution is or can be established from a packet capture

certificate chain, expiry and key strength are not observable for TLS 1.3 or resumed sessions (RFC 8446 SS2, SS2.2)

absence of a STARTTLS advertisement is ambiguous: stripping and genuine non-support are byte-identical at the application layer

NyaySaakshya cannot confirm that any remediation was applied or effective

13. Evidence and provenance

What was analysed, by which engine, and on what evidence.

Assessment identifier	2647f6ab59fca5e9
Capture identifier (SHA-256)	c2a84bf4ed973fcdff29bc5dd8ea24791d195fa186be3bbe7a359f68a3201fff
Analysis run identifier	33fbadb0e55244aa8a05eee2de96c9f0
Analysis timestamp	2026-10-03T04:36:38Z
Posture schema version	1.0
Posture engine version	0.8.0
Report schema version	1.0
Renderer version	0.9.0

Every conclusion in this report traces to specific frames in the capture, to the rule that examined them, and to the standard that rule cited. The capture identifier above is the SHA-256 of the analysed bytes; a capture that hashes differently is a different capture and this assessment does not describe it.

Engine provenance

ITEM	VALUE
Contradictions recorded	0
Duplicate sources merged	0
Note	every severity, status and standard originates in a Phase-4 or Phase-5 rule; the posture engine aggregates and never re-judges
Rule ids	['CS-STARTTLS-001', 'CS-STARTTLS-002', 'SEC-CERT-001', 'SEC-CFG-001', 'SEC-FS-001', 'SEC-KEX-001', 'SEC-PLAIN-002', 'SEC-STLS-001', 'SEC-STLS-002', 'SEC-TLS-001', 'SEC-TLS-002', 'SEC-TLS-003']
Source counts	{'abstentions': 7, 'cross_session_findings': 4, 'deterministic_findings': 11, 'fused_findings': 8, 'ml_results': 0}
Versions	{'posture_engine': '0.8.0', 'posture_schema': '1.0', 'scoring_formula': 'F2-group-damped'}

14. Methodology

The stages that produced this assessment.

Input	Packet capture (PCAP)
Protocols assessed	smtp
Output	Canonical posture assessment

1. Capture validation and SHA-256 identity
2. Packet dissection
3. TCP stream and protocol session reconstruction
4. Deterministic security analysis against versioned, standards-bound rules
5. Cross-session reasoning
6. Evidence fusion, deduplication and contradiction handling
7. Risk classification, posture scoring, prioritisation and remediation

The machine-learning lane was not exercised for this assessment.

Report schema 1.0 · Renderer 0.9.0 · Posture schema 1.0 · Posture engine 0.8.0

The timestamp shown is the time of analysis, not of report generation. This report is a rendering of the canonical assessment and introduces no security conclusion of its own.